MILITARY SPECIFICATION

INTEGRATED ELECTRONIC CENTRAL

AN/ASQ-56A

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope - The Electronic Central covered by this specification shall provide communication, navigation, and identification facilities required in naval aircraft to carry out military missions.

1.2 Classification - The Integrated Electronic Central shall include the following items:

| Item | Type Designation | MTBF | MIL-R-23094 Accept-Reject Figure |
|---|---|---|---|
| Receiver-Transmitter, Radio | RT-542(*)/ASQ | 170 | 13 |
| Receiver-Transmitter, Radio | RT-384(*)/ARN-52(V) | 250 | 13 |
| Coder-Receiver-Transmitter | KY-533/ASQ | 450 | 14 |
| Amplifier-Power Supply-Receiver | AM-2310(*)/ASQ | 400 | 14 |
| Antenna | AS-909(*)/ARA-48 | 2500 | 16 |
| Control, Radio Set (2) | C-4684/ASQ-56A | 5000 | 16 |
| Mounting-Plenum | MT-2243(*)/ASQ-56 | --- | -- |
| Adapter, Mounting | MX-4773(*)/ASQ-56A | --- | -- |
| Control, Radio Set (2) | C-2790(*)/ARC | 450 | 16 |
| Switching Unit | SA-629(*)/ARC | | |
| Mounting | MT-2141(*)/ARC | | |
| Indicator, Frequency Channel (2) | ID-758(*)/ARC | | |



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## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-E-4682 | Electron Tubes and Transistors, Choice and Application of |
| MIL-W-5088 | Wiring; Aircraft, Installation of |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-T-5422 | Testing, Environmental, Aircraft Electronic Equipment |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-C-6781 | Control Panel: Aircraft Equipment, Rack or Console Mounted |
| MIL-M-7793 | Meter, Time Totalizing |
| MIL-R-7893 | Radar Identification Set AN/APX-6B |
| MIL-C-7953 | Coder Group AN/APA-89 |
| MIL-I-9229 | Indicator, Course ID-387/ARN |
| MIL-E-17555 | Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of |
| MIL-T-18303 | Test Procedures; Preproduction and Inspection, Aircraft Electronic Equipment, Format for |
| MIL-N-18307 | Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment |
| MIL-R-18400 | Receiving Set, Radio AN/ARR-40 |
| MIL-E-19600 | Electronic Modules, General Specification for |



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Military

| | |
|---|---|
| MIL-N-22239 | Navigational Set, TACAN AN/ARN-52(V) |
| MIL-A-22669 | Amplifier-Power Supply-Receiver AM-2310/ASQ and Antenna AS-909/ARA-48 |
| MIL-C-22670 | Control, Radio Set C-2790/ARC; Indicator, Frequency Channel ID-758/ARC; Mounting MT-2141/ARC and Switching Unit SA-629/ARC |
| MIL-C-22672 | Coder-Receiver-Transmitter KY-533/ASQ |
| MIL-M-22673 | Mounting-Plenum MT-2243/ASQ-56 |
| MIL-R-22678 | Receiver-Transmitter, Radio RT-542/ASQ |
| MIL-E-23094 | Reliability Assurance for Production Acceptance of Avionics Equipment |
| MIL-A-81008 | Adapter, Mounting MX-4773/ASQ-56A |
| MIL-C-81009 | Control, Radio Set C-4684(*)/ASQ-56A |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization of |
| MS-17322( ) | Meter, Time Totalizing Miniature 400 Cycle, Digital |

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCI, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration.



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In so doing, the following shall govern:

(1) Parts and material requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (See 6.6.)

(3) When previously produced models of these centrals did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electronic tubes, transistors, and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The central shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction, and workmanship unless otherwise specified herein or in the individual equipment specifications.

3.2.2 Total Weight - The total weight of the central, excluding cables, shall be a minimum consistent with good design and shall not exceed 240.5 pounds.

3.2.3 Modular Construction - The use of modular space assignment and construction of easily disconnected plug-in subunits of component assemblies shall be in accord with Specification MIL-E-19600 where possible. The following considerations shall apply where practicable.

(1) Modules shall be of standardized form and dimensions to facilitate use in other equipment of different over-all configuration and shall be designed for maximum practical effectiveness of cooling air supplied by or to the equipment.



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(2) Each module shall contain the circuitry of one or more functions of the equipment to facilitate testing and repair. The equipment shall be divided into as many modular units as is electrically and physically practical in keeping with efficient space utilization, equipment performance, and reliability.

(3) It shall be possible to operate the equipment with one or more modules removed from the chassis but connected to it electrically by means of one or more extension cable assemblies and by use of mechnical extenders when required.

(4) It shall be possible to replace any module with other modules of the same type and obtain specified equipment performance with only minor readjustments.

3.2.4 Reliability

3.2.4.1 Operational Stability - The central shall operate with satisfactory performance for 50 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inacessible during flight.

3.2.4.2 Total Operating Life - The central shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.4.3 Time Totalizing Meter - A time totalizing meter conforming, as applicable, with Specification MIL-M-7793 and MS 17322( ) shall be included in the following:

RT-542/ASQ
RT-384/ARN-52(V)
KY-533/ASQ
AM-2310/ASQ

3.2.4.4 Reliability in Mean-Time-Between-Failures (MTBF) - The central shall have a minimum of 50 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3. For individual items when purchased separately the requirements shall be as shown in 1.2.

3.2.5 Cabling and Connections

3.2.5.1 Cables and Connectors - The central shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400 unless otherwise specified in the individual equipment specifications.



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3.2.5.2 Interconnection Cabling - The central shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be accomplished easily. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the centrals.

3.2.6 Soldering (Electrical Connections) - Secure mechanical connections shall not be required before soldering small lightweight parts to printed wiring assemblies provided central performance is not compromised.

3.2.7 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.2.8 Interference Control - The generation of radio interference by the central and the vulnerability of the central to radio interference shall be controlled within the limits of Specification MIL-I-6181 unless otherwise specified in the individual equipment specifications.

3.2.9 Maintenance provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.10 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307. (See 6.4.)

3.2.11 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

| | |
|---|---|
| Temperature | Room ambient ($25\pm5^{o}C$) |
| Altitude | Normal ground |
| Vibration | None |
| Humidity | Room ambient up to 90 percent relative humidity |
| Input Power Voltage | 115 $\pm$1.0 v a-c, 3 phase |
| | 26 $\pm$1.0 v a-c, 1 phase |
| | 27.5 $\pm$.5 v d-c |
| System Cooling Air | 132 lb per hour |

3.2.12 Service Conditions - The central shall operate satisfactorily under any of the environmental conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for the classes of equipment as specified herein unless otherwise specified in the individual equipment specification.



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| | | |
|---|---|---|
| Receiver-Transmitter, Radio | RT-542/ASQ | Class 2 |
| Receiver-Transmitter, Radio | RT-384/ARN-52(V) | Class 3 |
| Coder-Receiver-Transmitter | KY-533/ASQ | Class 2 |
| Amplifier-Power Supply-Receiver | AM-2310/ASQ | Class 2 |
| Antenna | AS-909/ARA-48 | Class 2 |
| Mounting-Plenum | MT-2243/ASQ-56 | Class 2 |
| Adapter-Mounting | 4773/ASQ-56A | Class 2 |
| Control, Radio Set | C-2790/ARC | Class 1 |
| Switching Unit | SA-629/ARC | Class 2 |
| Mounting | MT-2141/ARC | Class 2 |
| Indicator, Frequency Channel | ID-758/ARC | Class 1 |
| Control, Radio Set | C-4684/ASQ-56A | Class 1 |

3.2.12.1 Central Cooling Air Requirements - The central cooling air requirement is given in figure 1 of this specification.

3.2.12.2 Vibration - The central shall operate satisfactorily when subjected to the vibration requirements of curves I and II of Specification MIL-E-5400.

3.2.13 Warmup Time - The time required for the central to warm up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions, and 15 minutes at extreme service conditions.

3.2.14 Primary Input Power Requirements - The central shall meet all applicable requirements of MIL-STD-704 unless otherwise specified herein and shall give specified performance from the following power sources with characteristics as desined in MIL-STD-704 having limits as specified therein. The power required by the central shall not exceed the specified amounts.

(1) A-C Power, Three-Phase, 115/200V, Category "B," 900 VA.

(2) A-C Power, Single Phase, 26 V, Within Limits Proportional to Category "B," 125 VA.*

(3) D-C Power, 28 V, Category "B," 4.5 Amp Nominal.

(Additional current requirements of 6 to 8 amperes per equipment while channeling.)

3.2.15 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

*Includes typical system instrumentation.



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3.3 Detail Requirements

3.3.1 Communication Function - The communication function shall be performed by Receiver-Transmitter, Radio, RT-542/ASQ in accordance with Specification MIL-R-22678.

3.3.2 TACAN Function - The navigation function shall be performed by Receiver-Transmitter Radio RT-384/ARN-52(V) in accordance with Specification MIL-N-22239A.

3.3.3 Auxiliary Receiver, Power Supply and ADF Function - The auxiliary receiver, power supply, and automatic direction finder function shall be performed by Amplifier-Power Supply-Receiver AM-2310/ASQ and Antenna AS-909/ARA-48 in accordance with Specification MIL-A-22669.

3.3.4 IFF/Coder Function - The IFF/coder function shall be performed by the Coder-Receiver-Transmitter KY-533/ASQ in accordance with Specification MIL-C-22672

3.3.5 Communication Touch Tuning Control The communication touch tuning control with remote channel and frequency indicator function shall be performed by Control, Radio Set C-2790/ARC; Switching Unit SA-629/ARC; Mounting MT-2141/ARC and Indicator, Frequency Channel ID-758/ARC in accordance with Specification MIL-C-22670.

3.3.6 TACAN Control - The remote control function required for the operation of the TACAN shall be performed by Control, Radio Set C-4684/ASQ-56A in accordance with Specification MIL-C-81009.

3.3.7 Mounting-Plenum - The mounting-plenum function for the Integrated Electronic Central AN/ASQ-56A shall be performed by Mounting-Plenum MT-2243/ASQ-56A in accordance with Specification MIL-M-22673.

3.3.8 Mounting-Adapter - The mounting-adapter function for the Integrated Electronic Central AN/ASQ-56A shall be performed by Adapter, Mounting MX-4773/ASQ-56A in accordance with Specification MIL-A-81008.

3.4 Associated Equipment - The central shall operate with the following associated equipment which shall not be supplied as part of the central.

3.4.1 Auxiliary Receiver Control - The remote control function required for the operation of the auxiliary receiver shall be performed by Control, Receiver C-1457/ARR-40 of Receiving Set, Radio AN/ARR-40 in accordance with Specification MIL-R-18400.

3.4.2 IFF Control - The remote control function required for the operation of the IFF shall be performed by the Control, Radar Set C-1159/APX-6B of Radar Identification Set AN/APX-6B in accordance with Specification MIL-R-7893.



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3.4.3 Coder Control - The remote control function required for the operation of the Coder shall be performed by Control, Radar Set C-1272A/APA-89 of Coder Group AN/APA-89 in accordance with Specification MIL-C-7953.

3.4.4 TACAN Indicator - The indicator function required for the remote operation of TACAN shall be performed by Indicator, Course ID-387/ARN in accordance with Specification MIL-I-9229.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform only at the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more individual equipments and centrals representative of production equipments and centrals (see 4.2.3) to be supplied under the contract. Preproduction tests shall consists of the following:

(1) Contractor's Demonstration Test

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipments to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.



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4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts, these data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating, and the applied circuit voltage, current, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the individual equipments and/or central shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipments submitted for service approval tests, the contractor also shall submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipments and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include interference tests in accordance with Specification MIL-I-6181 and environmental tests in accordance with MIL-T-5422. Tests will be on a central basis only.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Centrals supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality be equivalent to the approved preproduction sample. Each central shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection, and for equipment already accepted by the government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.



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4.3 Production Sample Tests - When requested by the procuring activity, one central shall be selected from the first 10 production centrals submitted for acceptance and sent to a designated government laboratory for tests. The equipments shall be selected by the government inspector after the equipments have passed all individual tests successfully.

4.3.1 Scope of Tests - These equipments may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipments are equivalent to the previously approved preproduction samples in design, construction, and performance and that they meet all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished on equipments previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test, the equipments shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipments for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests of individual equipments of the systems shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests



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4.4.1 Individual Tests - Each individual equipment submitted for acceptance shall be subjected to the individual tests except that one out of every ten shall be tested on a central basis only. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

| | |
|---|---|
| Temperature | Ambient room |
| Humidity | Ambient room |
| Vibration (10 minutes) | Any selected frequency within the range of 20 to 20 cps (excluding resonant points) and a minimum amplitude of $\pm$3 g |

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes, prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period



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need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on centrals which have successfully passed the individual tests. The following centrals shall be selected by the government inspector for sampling tests: (Sampling tests shall not be conducted unless reliability tests of 4.4.3 are deleted by contractual action.)

| Quantity of Centrals Offered for Acceptance | Quantity to be tested |
|---|---|
| First 10 | 1 |
| Next 50 | 1 |
| | 1 for each additional 100 or fraction thereof |

4.4.2.1 Scope of Tests - As a minimum, each central selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each central tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed concerning cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Unless otherwise stipulated in the contract, equipments which have successfully passed the individual tests shall be tested for reliability assurance in accordance with Specification MIL-R-23094. (Procedure II).

4.4.3.1 Test Level - The test level shall be level III (from table I of Specification MIL-R-23094).



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4.4.3.2 Accept-Reject Criteria - Figure 13 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria. MTBF specified in 3.2.4.4 or referenced specication apply.

4.4.3.3 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 4 hours during each cycle.

4.4.3.4 Performance Characteristics to be Measured

(1) UHF Communications - RT-542/ASQ

| Characteristic | Specification Requirement | Acceptance Requirement (if lower) |
|---|---|---|
| Receiver Sensitivity | Standard 5 vu/10 db<br>Service 20 uv/10 db | 20 uv/10 db |
| Audio Capability | Standard not less than 250 mw | |
| | Service not less than 67.5 mw | Not less than 67.5 mw |
| R-F Power Output | Standard not less than 16 w | Not less than 9 w |
| | Service not less than 9 w | |
| Modulation Capability (input to clipping) | Not less than 80% | Not less than 75% |
| Tone Modulation Capability | 70 - 100% | 70 - 100% |
| Side-Tone Level | Standard not less than 250 mw | Not less than 75 mw |
| | Service not less than 75 mw | |



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(2) Receiver/ADF/Power Supply - AM-2310/ASQ and AS-909/ARA-48

| Characteristic | Specification Requirement | Acceptance Requirement (if lower) |
|---|---|---|
| Receiver Sensitivity | Standard 6 uv/10 db<br>Service 18 uv/10 db | 18 uv/10 db |
| Audio Capability | Standard not less than 250 mw<br><br>Service not less than 67.5 mw | Not less than 67.5 mw |
| Power Supply Voltages | | |
| 430 v d-c | 370 to 505 | 360 to 515 |
| 275 v d-c | 241 to 302 | 230 to 310 |
| 250 v d-c | 218 to 278 | 210 to 290 |
| 130 v d-c | 114 to 142 | 110 to 145 |
| 140 v d-c | 137 to 143 | 135 to 145 |
| -180 v d-c | -148 to -207 | -140 to -215 |
| ADF 100-CPS Inverter | Not less than 2.7 v a-c | Not less than 2.7 v a-c |
| ADF Motor Control | Not less than 75 v a-c | Not less than 75 v a-c |
| ADF R-F Operational Test | Not more than 6 seconds for antenna to assume new heading 180° removed from previous heading | Not more than 12 seconds |

(3) TACAN RT-384/ARN-52(V)

| Characteristic | Specification Requirement | Acceptance Requirement (if lower) |
|---|---|---|
| Distance Accuracy and Sensitivity | With sig -87 dbm to 0 dbm, ±0.1 mi ±0.2% range 0 to 50 mi and ±0.2 mi ±.033% range 50 mi to max. | With sig -87 dbm to 0 dbm, ±0.1 mi ±0.2% range 0 to 50 mi and ±0.2 mi ±.033% range 50 mi to max. |



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| Characteristic | Specification Requirement | Acceptance Requirement (if lower) |
|---|---|---|
| Bearing Accuracy and Sensitivity | ±2°, -80 dbm to -77 dbm; ±1°, -77 dbm to -5 dbm; ±2°, -5 dbm to 0 dbm. | ±2°, -80 dbm to -77 dbm ±1°, -77 dbm to -5 dbm ±2°, -5 dbm to 0 dbm. |
| Interrogation Coding | 12 ±0.5 usec | 12 ±0.5 usec |
| Identification | Not less than 50 mw audio for sig levels from -87 to 0 dbm. | Not less than 50 mw audio for sig level from -87 to 0 dbm. |
| Power Output | Not less than 1.2 kw Not more than 3.0 kw | Not less than1.2 kw Not more than 3.0 kw |

(4) IFF/Coder KY-533/ASQ

| Characteristic | Specification Requirement | Acceptance Requirement (if lower) |
|---|---|---|
| Receiver Center Frequency | 1030 ±.5 mc | 1030 ±1.0 mc |
| Normal Receiver Triggering Level | 83 dbv | 80 dbv |
| Interrogation Characteristics | | |
| Mode 1 | 3 ±.2 usec | 3 ±.2 usec |
| Mode 2 | 5 ±.2 usec | 5 ±.2 usec |
| Mode 3 | 8 ±.2 usec | 8 ±.2 usec |
| Transmitter Frequency | 1090 ±3 mc | 1090 ±3 mc |
| SIF Reply Pulse Characteristics | | |
| Bracket Pulses | 20.30 ±0.1 usec | 20.30 ±0.1 usec |
| Mode 1 (5 pulses at 2.0-usec intervals) | ±.05 usec | ±.05 usec |
| Mode 2 (12 pulses at 1.45-usec intervals) | ±.05 usec | ±.05 usec |
| Mode 3 (6 pulses at 2.9-usec intervals) | ±.05 usec | ±.05 usec |
| Transmitter Power | 27±3 dbw | NLT 24 dbw |



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4.4.3.5 Failure Criteria - Performance requirements below those specified herein (4.4.3.4) shall constitute one failure unless otherwise specified in the approved detailed test procedures (see 4.6). If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused other parts to fail.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Tests Schedule - Selection of individual equipments and centrals for special tests shall be made as follows:

(1) On an early equipment after an engineering change.

(2) Whenever failure reports or other information indicate additional equipments should be tested

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken.

(1) Determine the cause of failure.

(2) Determine whether the failure is an isolated case or design defect.

(3) Submit to the procuring activity proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been corrected satisfactorily.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 500 hours duration and shall be conducted on centrals that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with



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the following; except that when a production sample equipment is requested and submitted to the procuring activity for test, then the first equipment specified below need not be selected or subjected to this test. (Centrals which have passed the sampling tests or special tests successfully may be selected for life tests.)

| Quantity of Centrals Offered for Acceptance | Quantity to be Selected for Life Test |
|---|---|
| First 25 | 1 |
| Next 175 | 1 |
| Next 300 | 1 |
| | 1 for each additional 500 or fraction thereof |

4.5.1 Test Conditions - The central life test shall be conducted under the following simulated service conditions:

| | |
|---|---|
| Temperature | Normal Room |
| Altitude | Normal ground (0-5000 ft) |
| Humidity | Room ambient |
| A-C Voltage | 115 ±5 volts (at lowest applicable frequency) |
| D-C Voltage | 27.0 ±2.0 volts |
| Cooling Air | 132 lb per hour |

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stablilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment or central, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part number

(2) The circuit reference symbol-number

(3) The part function



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(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - A central which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment or central for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts, such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been tested and inspected previously by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.



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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The Electronic Central shall provide communication, navigation, and identification facilities required in naval aircraft to carry out military missions.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5±0.5 v d-c, 26±1.0 v a-c, and 115 ±1.0 a a-c. These data are to be used in testing the systems at installation points for compliance with minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviations from this specification where applicable shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply only to the extent specified.

6.4 The parentheses(*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this system. Where it appears substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation



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shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. . Each request shall be accompanied by complete supporting information.

6.6 As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With a few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).



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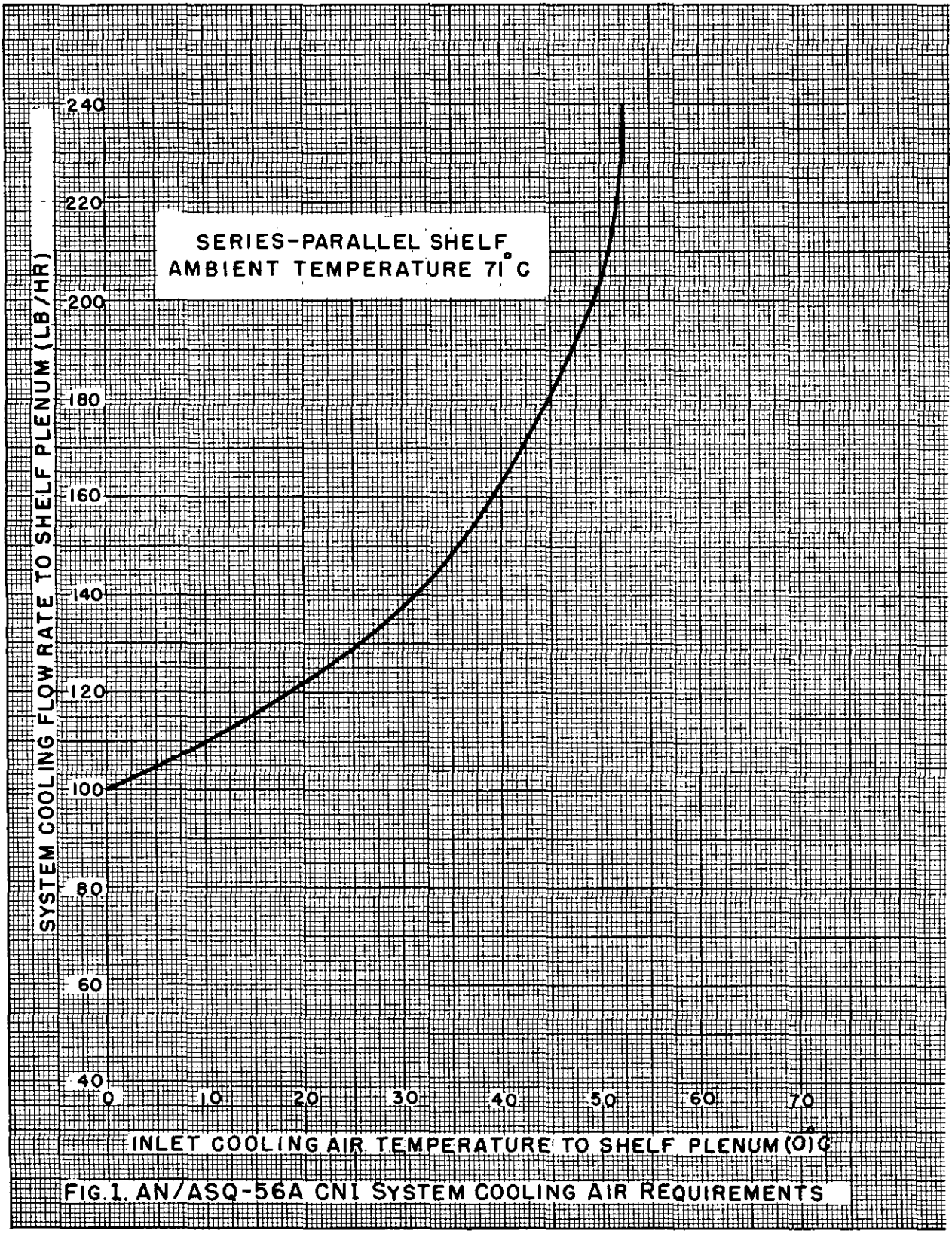



FIG.1. AN/ASQ-56A CNI SYSTEM COOLING AIR REQUIREMENTS



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-81027(Navy) Integrated Electronic Centr-l AN/ASQ-56A

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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